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MID TERM EXAMINATION

B. TECH PROGRAMMES (UNDER THE AEGIS OF USICT)
Second Semester, May, 2023

Paper Code: BS 106**Subject: Applied Physics-II****Time: 1.5 Hours****Max. Marks: 30****Note: Attempt Question no. 1 which is compulsory and any two more questions from remaining.**

Q. No.	Question	Marks	COs
Question - 1			
1 (a)	Distribute three particles in three energy states in accordance with MB, BE and FD statistics.	2.5	CO2
1 (b)	What do you understand by normalisation of a wave function? Give the essential features of an acceptable wave function.	2.5	CO1
1 (c)	Determine the uncertainty in position of an electron whose speed is measured to be 5.0×10^3 m/s to an accuracy of 0.003%.	2.5	CO1
1 (d)	Compute the root mean square speed of oxygen molecules at 0°C. Given mass of $O_2 = 5.31 \times 10^{-26}$ kg.	2.5	CO2
Question - 2			
2 (a)	Evaluate the expectation value of the position and the momentum of a particle trapped in a box of width L.	5	CO1
2 (b)	Determine the energy required to jump from ground state to second excited state by an electron trapped in a box of length 1 Å.	2	CO1
2 (c)	Brief the conditions required for quantum tunnelling. Calculate the transmission probability for an electron of energy 2 eV when incident upon a rectangular potential barrier of height 6 eV and width 0.50 nm.	3	CO1
Question - 3			
3 (a)	Find the expression of the average molecular energy of an ideal gas molecule.	4	CO2
3 (b)	How do Fermions differ from Bosons? Give examples.	3	CO2
3 (c)	Calculate the average energy and speed of a free electron at 0 K for a given substance. Given Fermi energy of substance is 7.9 eV.	3	CO2
Question - 4			
4 (a)	Using Schrödinger's equation, find the wave function for a particle of mass m in an infinite potential well of width L. Also find its eigen energy values.	6	CO1
4 (b)	An electron gas obeys the Maxwell-Boltzmann statistics. Calculate the average thermal energy (in eV) for an electron in the system at 300 K.	2	CO2
4 (c)	State Planck's radiation law and give its expression for energy density.	2	CO2

****** End of Question Paper ******